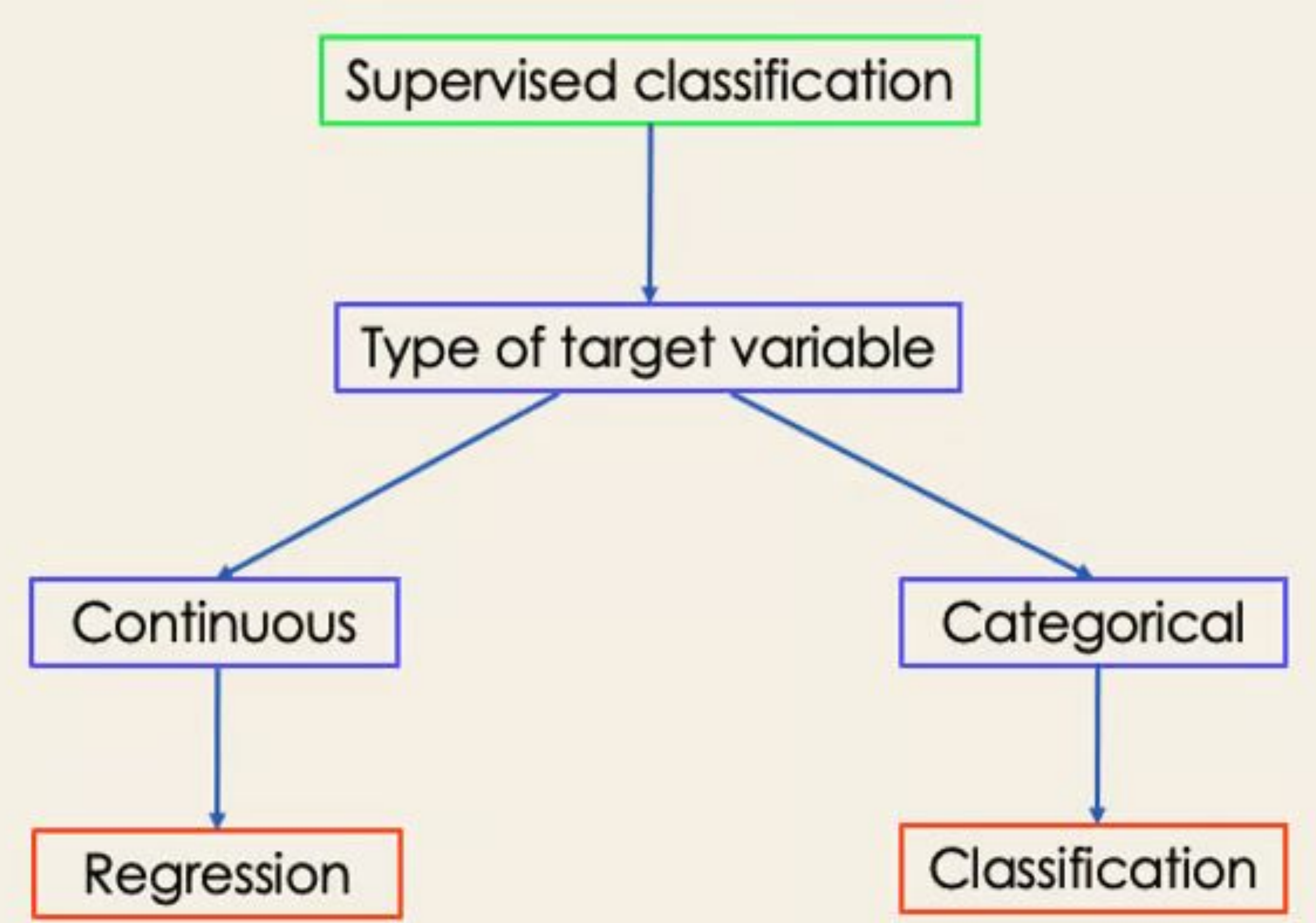
Classification Algorithm in Machine Learning

Supervised Machine Learning algorithm

- Supervised Machine Learning algorithm can be broadly classified into Regression and Classification Algorithms. In Regression algorithms, we have predicted the output for continuous values, but to predict the categorical values, we need Classification algorithms.


Supervised classification
Type of target variable
Continuous
Categorical
Regression
Classification

What is the Classification Algorithm

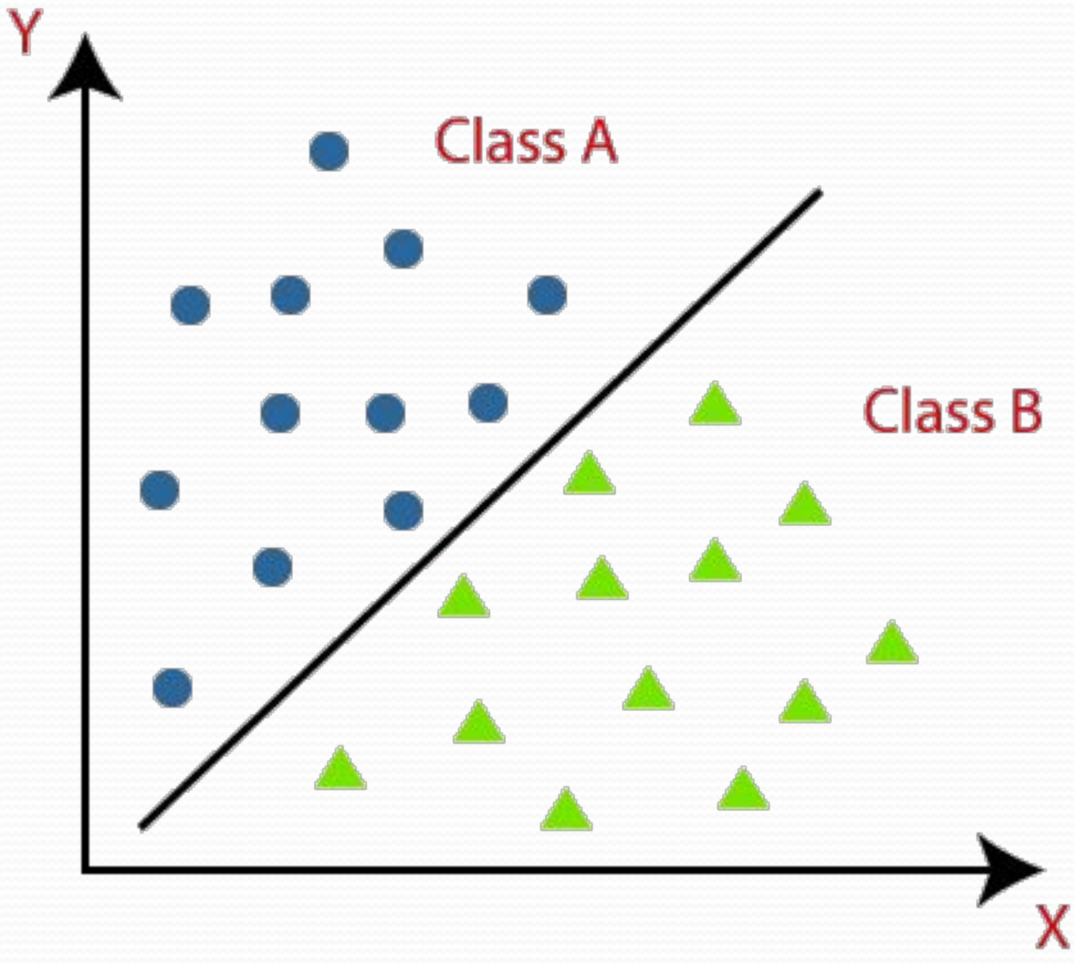
- The Classification algorithm is a Supervised Learning technique that is used to identify the category of new observations on the basis of training data.
- In Classification, a program learns from the given dataset or observations and then classifies new observation into a number of classes or groups.
- Such as, **Yes or No, 0 or 1, Spam or Not Spam, cat or dog**, etc. Classes can be called as targets/labels or categories.

Cont…

- Unlike regression, the output variable of Classification is a category, not a value, such as "Green or Blue", "fruit or animal", etc.
- Since the Classification algorithm is a Supervised learning technique, hence it takes labeled input data, which means it contains input with the corresponding output.
- y=f(x), where y = categorical output
- The best example of an ML classification algorithm is **Email Spam Detector**.

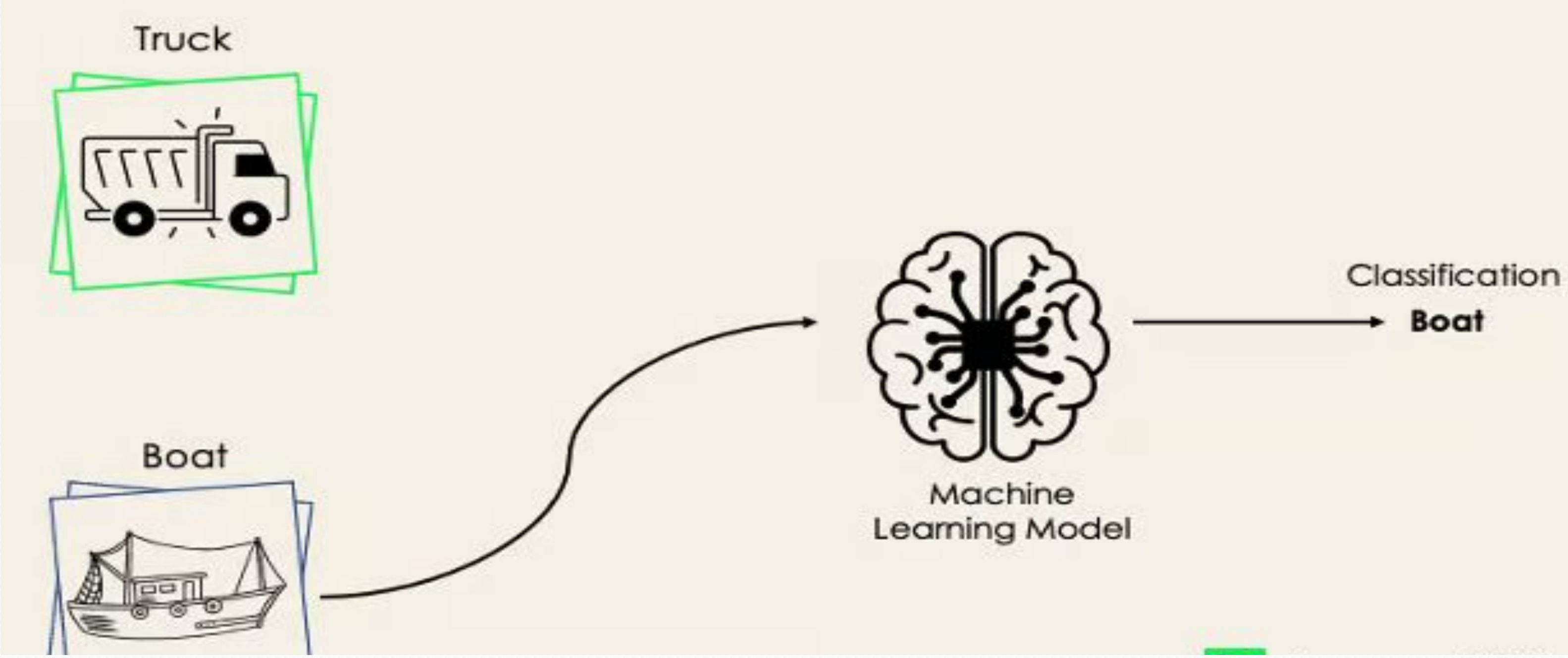
Classification

- The main goal of the Classification algorithm is to identify the category of a given dataset, and these algorithms are mainly used to predict the output for the categorical data.
- Classification algorithms can be better understood using the below diagram. In the below diagram, there are two classes, class A and Class B. These classes have features that are similar to each other and dissimilar to other classes.


Y
Class A
Class B
X

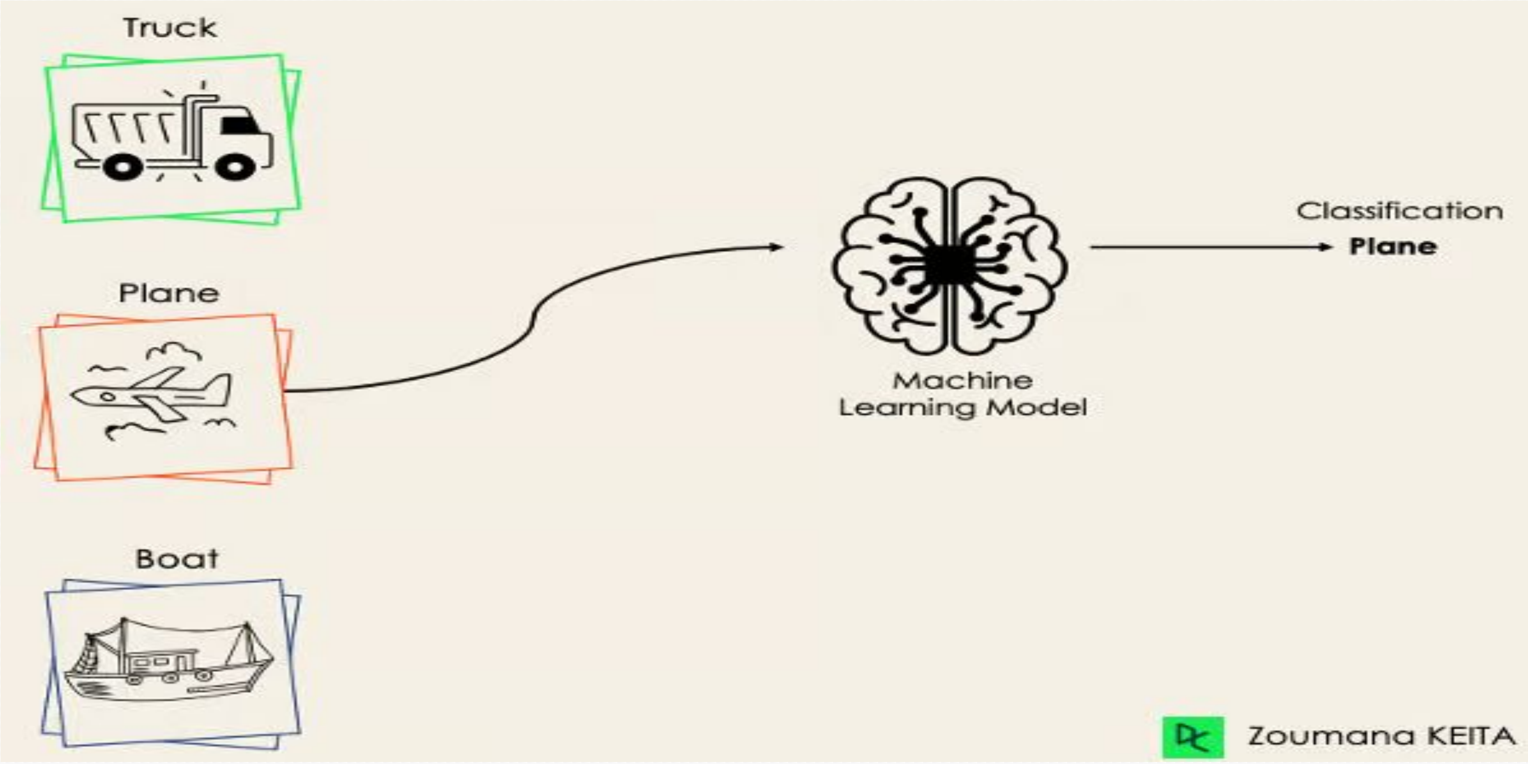
There are two types of Classifications:

- **Binary Classifier:** If the classification problem has only two possible outcomes, then it is called as Binary Classifier.
  Examples: YES or NO, MALE or FEMALE, SPAM or NOT SPAM, CAT or DOG, etc.



Multi-class Classifier

- **Multi-class Classifier:** If a classification problem has more than two outcomes, then it is called as Multi-class Classifier. **Example:** Classifications of types of crops, Classification of types of music.



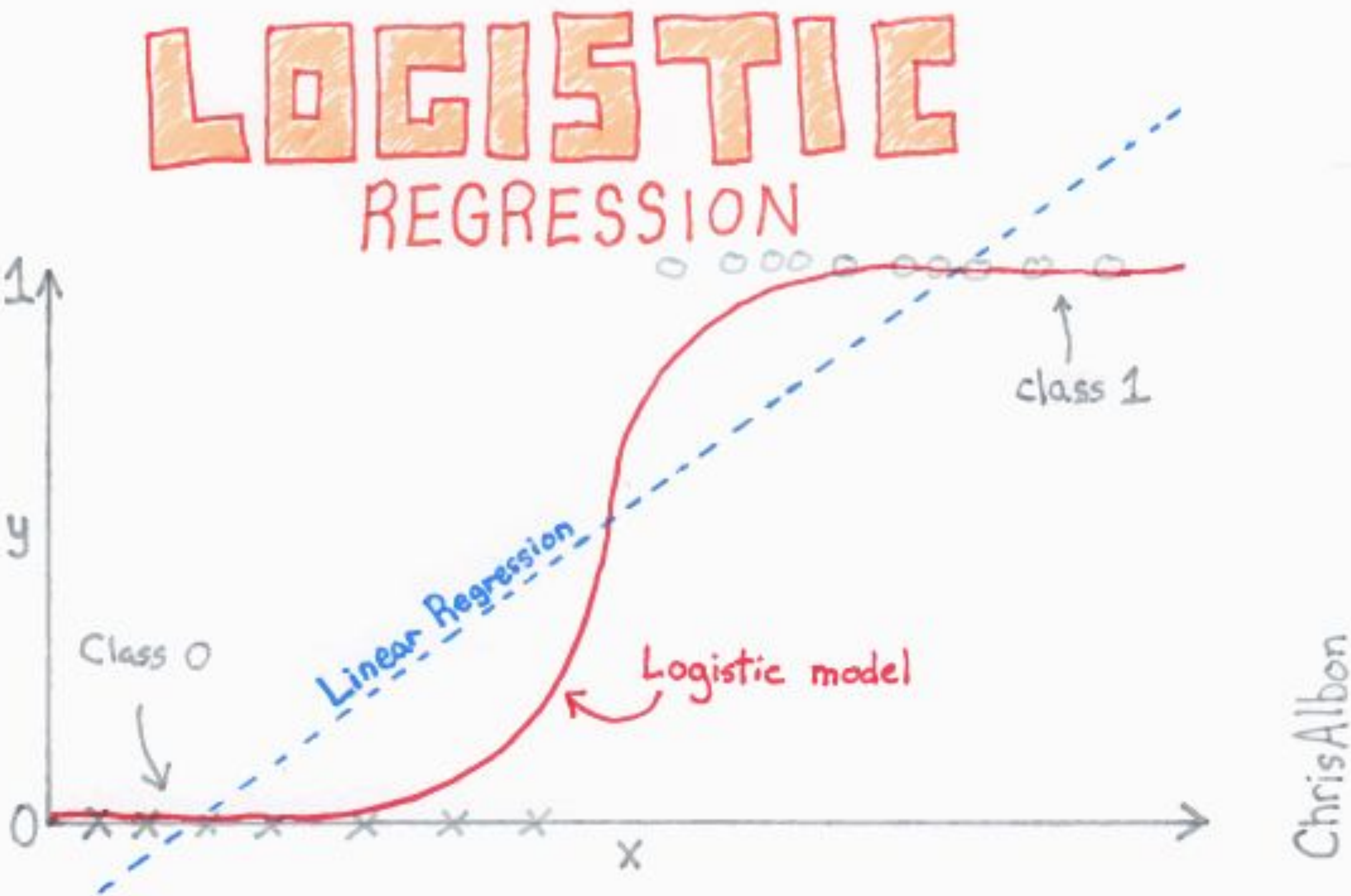
Types of ML Classification Algorithms:

- Classification Algorithms can be further divided into the Mainly two category:
- **Linear Models**
  - Logistic Regression
  - Support Vector Machines
- **Non-linear Models**
  - K-Nearest Neighbours
  - Kernel SVM
  - Naïve Bayes
  - Decision Tree Classification
  - Random Forest Classification

Logistic Regression

- It is a classification algorithm in machine learning that uses one or more independent variables to determine an outcome. The outcome is measured with **have only two possible outcomes**.
- The goal of logistic regression is to find a best-fitting relationship between the dependent variable and a set of independent variables.

Logistic regression is kind of like linear regression, but is used when the dependent variable is not a number but something else (e.g., a "yes/no" response). It's called regression but performs classification based on the regression and it classifies the dependent variable into either of the classes.


LOGISTIC
REGRESSION
1
0
y
x
class 1
Class 0
Linear Regression
Logistic model
ChrisAlbon

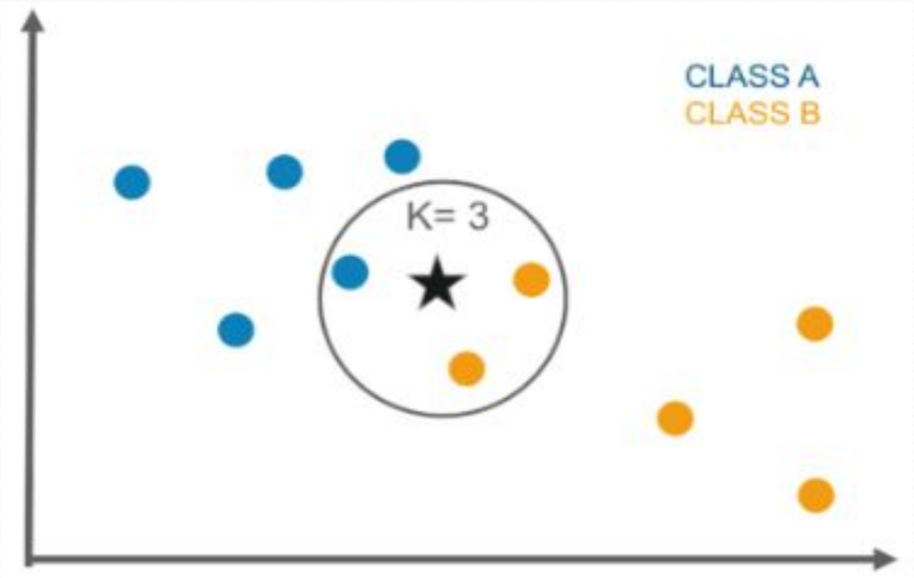
Naive Bayes Classifier

- It is a classification algorithm based on **Bayes's theorem** which gives an assumption of independence among predictors.
- Naive Bayes model is easy to make and is particularly useful for comparatively large data sets.
- The Naive Bayes classifier requires a small amount of training data to estimate the necessary parameters to get the results. They are extremely fast in nature

$$P(C_i|\, x_1, x_2 \ldots, x_n) = \frac{P(x_1, x_2 \ldots, x_n|C_i).P(C_i)}{P(x_1, x_2 \ldots, x_n)} \; for\; 1 < i < k$$

K-Nearest Neighbor

- It is a algorithm that **stores all instances corresponding to training data in n-dimensional space**. It is a **lazy learning algorithm** as it does not focus on constructing a general internal model, instead, it works on storing instances of training data.



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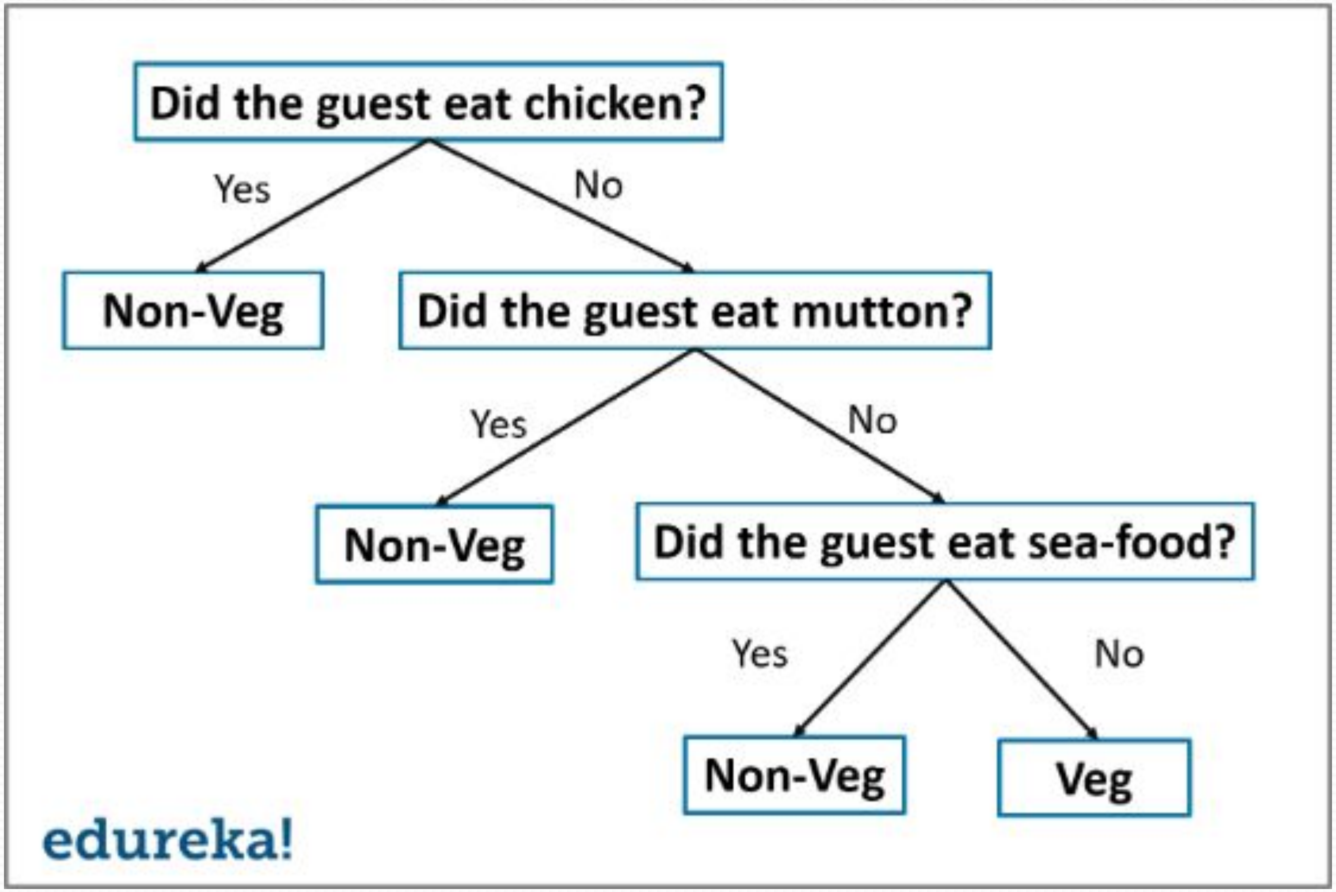
- Classification is computed from a simple majority vote of the k nearest neighbors of each point.
- It is supervised and takes a bunch of labeled points and uses them to label other points.
- To label a new point, it looks at the labeled points closest to that new point also known as its nearest neighbors.

Decision Tree

- The decision tree algorithm builds the classification model in the form of a **tree structure**.
- It utilizes the if-then rules which are equally exhaustive and mutually exclusive in classification. The process goes on with breaking down the data into smaller structures and eventually associating it with an incremental decision tree
- The final structure looks like a tree with nodes and leaves. The **rules are learned sequentially** using the training data one at a time

Decision Tree



Evaluating a Classification model:

- Once our model is completed, it is necessary to evaluate its performance; either it is a Classification or Regression model.
- So for evaluating a Classification model, we have the following ways:
- **1. Log Loss or Cross-Entropy Loss**
- **2.Confusion Matrix**
- **3. AUC-ROC curve:**

Log Loss or Cross-Entropy Loss

- It is used for evaluating the performance of a classifier, whose output is a probability value between the 0 and 1.
- For a good binary Classification model, the value of log loss should be near to 0.
- The value of log loss increases if the predicted value deviates from the actual value.
- The lower log loss represents the higher accuracy of the model.
- For Binary classification, cross-entropy can be calculated as:

Confusion Matrix

- The confusion matrix provides us a matrix/table as output and describes the performance of the model.
- It is also known as the error matrix. The matrix consists of predictions result in a summarized form, which has a total number of correct predictions and incorrect predictions.

	Actual Positive	Actual Negative
Predicted Positive	True Positive	False Positive
Predicted Negative	False Negative	True Negative

$$\text{Accuracy} = \frac{TP+TN}{\text{Total Population}}$$

AUC-ROC curve

- ROC curve stands for **Receiver Operating Characteristics Curve** and AUC stands for **Area Under the Curve**.
- It is a graph that shows the performance of the classification model at different thresholds.
- To visualize the performance of the multi-class classification model, we use the AUC-ROC Curve.
- The ROC curve is plotted with TPR and FPR, where TPR (True Positive Rate) on Y-axis and FPR(False Positive Rate) on X-axis.

A bit of context
Imagine you are a healthcare startup, and want an AI assistant able to predict whether a given patient has a heart disease or not based on its health record. This is a binary classification problem where the model will predict

- 1, True or Yes if the patient has heart disease
- 0, False or No otherwise

1 Confusion matrix

A 2X2 matrix that nicely summarizes the number of correct predictions of the model. It also helps in computing different other performance metrics.

Predicti / Reality	Yes	No
Yes	True Positives (TP)	False Negatives (FN)
No	False Positives(FP)	True Negatives (TN)

Type I Error

Type II Error

Type I & II Errors can be used interchangeably when referring to False Positives and False negatives respectively

2 Accuracy

We get *accuracy* by answering this question: **"out of the predictions made by the model, what percentage is correct?"**

$$Accuracy = \frac{TP + TN}{Total\ number\ observation}$$

3 Precision

We get <u>precision</u> by answering this question: **"out of all the YES predictions, how many of them were correct?"**

$$\textit{Precision} = \frac{TP}{TP + FP}$$

4 Recall / Sensitivity

It aims to answer this question: **"how good was the model at predicting <u>real Yes events</u>?"**, which can be considered as the flip of the precision.

$$\textit{Sensitivity} = \frac{TP}{TP + FN}$$

5 Recall / Specificity

It aims to answer this question: **"how good was the model at predicting <u>real No events</u>?".**

$$\textit{Specificity} = \frac{TN}{TN + FP}$$

6 F1 Score

Sometimes used when dealing with imbalanced data set, meaning that there are more of one class/label than there are of the other. It corresponds to the harmonic mean of the precision and recall.

$$\textit{F1 Score} = 2 \, . \, \frac{\textit{Precision x Recall}}{\textit{Precision + Recall}}$$

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7 AUC – ROC Curve

AUC– ROC generates probability values instead of binary 0/1 values. It should be used when your data set is roughly balanced.

Using ROC for imbalanced data sets lead to incorrect interpretation.

ROC curves provide good overview of trade-off between the TP rate and FP rate for binary classifier using different probability thresholds.

- A value below 0.5 indicates a poor classifier
- A value of 0.5 means random classifier
- Value over 0.7 corresponds to a good classifier
- 0.8 indicates a strong classifier
- We have 1 when the classifier perfectly predicts everything.

Strategies to choose the right metric

Choose accuracy
- The cost of FP and FN are roughly equal.
- The benefit of TP and TN are roughly equal.

Choose Precision
- The cost of FP is much higher than a FN.
- The benefit of a TP is much higher than a TN.

Choose recall
- The cost of FN is much higher than a FP.
- The cost of a TN is much higher than a TP.

ROC AUC & Precision – Recall curves
- Use ROC when the dealing with balanced data sets.
- Use precision-recall for imbalanced data sets.

Use cases of Classification Algorithms

- Classification algorithms can be used in different places. Below are some popular use cases of Classification Algorithms:
- Email Spam Detection
- Speech Recognition
- Identifications of Cancer tumor cells.
- Drugs Classification
- Biometric Identification, etc.